

Siddharth Bayapureddy

+91 7075286119 | siddharthbayapureddy@gmail.com | siddharth-bayapureddy.vercel.app | linkedin.com/in/siddharth-bayapureddy | github.com/SiddharthBayapureddy

SUMMARY

Machine learning engineer with hands-on experience building end-to-end MLOps pipelines, generative AI systems, and LLM-powered agentic applications. Proficient in PyTorch, AWS SageMaker, retrieval-augmented generation (RAG), and few-shot learning. Seeking AI/ML engineering or applied research internships where strong fundamentals meet production-grade thinking.

EDUCATION

BITS Pilani, Hyderabad Campus

Bachelor of Engineering in Computer Science (CGPA: 9.1/10)

Hyderabad, TS

Aug. 2024 – Jul. 2028

PROJECTS

Vigil | *Python, XGBoost, AWS Lambda, Step Functions, SageMaker, Docker, Mistral*

Jun. 2026 – Present

- Engineered an autonomous MLOps pipeline for aircraft engine predictive maintenance (NASA CMAPSS FD001); PSI/KS drift detection triggers a SageMaker-backed self-healing retraining loop, eliminating all manual intervention
- Boosted predictive accuracy by 58% (R^2 : 0.57 $\rightarrow$ 0.90, MAE: 9.3 cycles) via 112-feature rolling window engineering; orchestrated zero-downtime champion/challenger promotion via AWS Step Functions (SageMaker retraining $\rightarrow$ validation $\rightarrow$ live swap) with Mistral LLM drift narration

SparseSynth | *Python, Stable Diffusion, LoRA, Meta-Learning, PyTorch, Transformers*

Sep. 2025 – Jan. 2026

- Built a meta-learning pipeline that synthesizes 500+ training samples from 30–40 references ($13\times$ augmentation ratio), solving low-data constraints without manual annotation
- Designed a Transformer-based aggregator to predict task-specific LoRA weights and injected them into a Stable Diffusion backbone, enabling few-shot generalization to unseen concepts at inference time

Vorp | *Python, LiteLLM, RAG, ChromaDB, Typer, Rich, PyPI*

Dec. 2025 – Jan. 2026

- Built and published a terminal-based AI agent on PyPI for natural-language file operations, codebase Q&A, and developer workflow automation
- Routed across multiple LLMs via LiteLLM based on task complexity; integrated a ChromaDB-backed RAG pipeline for semantic retrieval over local codebases with persistent conversation history

CLUBS & LEADERSHIP

ACM – GenAI & ML

Sep. 2025 – Present

BITS Pilani, Hyderabad Campus

- Built a real-time face anonymizer using flow matching, achieving lower-latency inference than diffusion-based baselines by reducing required sampling steps
- Developed a resume screening tool that maps student profiles to PS-II and placement trends, generating data-driven, actionable feedback

Lynq – Agentic AI

Sep. 2025 – Present

BITS Pilani, Hyderabad Campus

- Built an agentic expense-validation system for campus events that cross-checks submitted bills against verified purchases and flags discrepancies automatically
- Integrated Crawl4AI to scrape real-time market prices and compare them against submitted receipts, automating pricing-mismatch detection

TECHNICAL SKILLS

Languages: Python, Java, C, C++, SQL

ML & AI: PyTorch, scikit-learn, XGBoost, Hugging Face Transformers, LangChain, LangGraph, LiteLLM, Stable Diffusion, LoRA, NumPy, Pandas, Matplotlib

MLOps & Cloud: Amazon Web Services (AWS SageMaker, Lambda, Step Functions), Docker, Git, GitHub, Jupyter Notebooks

Backend & Developer Tools: FastAPI, Typer, Rich, Postman, Crawl4AI

Databases: PostgreSQL, ChromaDB, Supabase